

★ PRIMER · UNIT 3

Before the Lecture: What Are We Predicting?

Unit 3 · Labeling and Meta-Labeling

Watch this first if precision and recall are new

Twenty minutes of groundwork — what a label is, how a confusion matrix works, and why 'will it go up?' is the wrong question to ask a market.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Precision and recall are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 3 Sits

The course builds one pipeline; here is this unit's stone

Unit 2

done

Bars, cleaning, and log returns.

Unit 3

you are here

Fix the label — what are we predicting?

Unit 4

next

Linear models and regularization.

The best model in the world cannot rescue a badly chosen answer key.

What a Label Really Is

A label is the answer key. Choose it badly and the model learns the wrong lesson perfectly.

Where it comes from

- You define it, not the data.
- It encodes what you care about.
- Different label, different model.

Why it is hard here

- Future prices are not answers.
- Timing and size both matter.
- Trades overlap in time.

Unit 3 is the only unit about the answer key. Everything after assumes it is right.

The Words the Lecture Assumes

Five terms from the confusion matrix, and the lecture opens up

Word	What it means	Trading version
True positive	you said yes, and it was yes	you traded, and won
False positive	you said yes, it was no	you traded, and lost
Precision	of your yeses, how many were right	hit rate of your trades
Recall	of all the yeses, how many you caught	opportunities not missed
Imbalance	one class far rarer than the other	defaults are 3% of loans

Precision is about the trades you took. Recall is about the ones you did not.

Precision and Recall Pull Apart

You cannot maximise both — the lecture makes you choose

If you	Precision	Recall	Trading meaning
trade everything	low	high	you catch all, and bleed fees
trade only sure things	high	low	you win often, and rarely act
trade nothing	undefined	zero	no losses, no business

Meta-labeling exists precisely to buy precision without giving up all the recall.

Why 'Up Tomorrow?' Is the Lazy Question

A fixed horizon ignores how far the price moved, and whether you would have been stopped out first.

What it misses

- A 0.01% rise counts as up.
- A crash mid-week is invisible.
- Volatility is ignored.

What a trader asks

- Did I hit my target first?
- Or my stop first?
- Or neither, and time ran out?

Three finish lines instead of a calendar. That is the triple barrier, in one sentence.

A Confusion Matrix by Hand

Ten signals, six of them acted on — fill in the four cells yourself

Signal	You acted?	Outcome	Cell
s1–s3	yes	won	true positive × 3
s4–s5	yes	lost	false positive × 2
s6	yes	won	true positive × 1
s7–s8	no	would have won	false negative × 2
s9–s10	no	would have lost	true negative × 2

Four true positives, two false. Precision = $4/6 = 67\%$. Recall = $4/6 = 67\%$.

Reading Those Two Numbers

Same data, two questions — and they answer different worries

Precision = 4 / 6

- Of six trades, four won.
- This is your hit rate.
- It is about wasted capital.

Recall = 4 / 6

- Of six chances, you took four.
- This is your coverage.
- It is about missed profit.

Meta-labeling keeps the signal and learns a second model for whether to act on it

Why Class Imbalance Breaks Accuracy

The single most quoted number is also the most useless one here

A 97% accurate model

- Defaults are 3% of loans.
- Predict 'no default' always.
- You score 97%.

And it is worthless

- It never finds a default.
- Accuracy hid that completely.
- Precision and recall do not.

Whenever someone quotes accuracy on rare events, ask what the base rate was.

How a Label Gets Made

Three moves, and the lecture spends a slide on each

1 · Set the bars

target and stop

Two horizontal lines, scaled to the stock's own volatility.

2 · Add a clock

the third barrier

If neither line is hit in N days, the trade times out.

3 · Read the label

which came first

Win, loss, or timeout — that is what the model learns.

The bars must scale with the stock. A fixed 2% means nothing across two names.

Three Ways a Fixed Horizon Lies

These are the lecture's own three examples

The lie	What happens	Cost to you
Ignores size	+0.01% labelled the same as +8%	you chase noise
Ignores path	a crash mid-week is invisible	you are stopped out
Ignores volatility	2% means different things	labels are incomparable

Each lie is silent — the dataset looks perfectly reasonable while carrying it.

Notation You Will Meet

Four symbols carry the labeling section

Symbol	Read it as	It stands for
pt	profit taking	the upper barrier, the target
sl	stop loss	the lower barrier
y	why	the label — win, loss or timeout
σ	sigma	volatility, which scales the bars

The barriers are written as multiples of sigma, never as fixed percentages.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>df['ret'].rolling(20).std()</code>	rolling volatility, to scale the barriers
<code>np.sign(x)</code>	turn a number into -1 , 0 or $+1$
<code>confusion_matrix(y, yhat)</code>	the four cells you just filled in by hand
<code>precision_score(y, yhat)</code>	of your yeses, how many were right
<code>recall_score(y, yhat)</code>	of all the yeses, how many you caught

Two lines of scikit-learn replace the table — but only after you can draw it.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 4

1

"Accuracy is the score that matters"

On rare events accuracy rewards doing nothing. Precision and recall are the honest pair.

2

"The label is given by the data"

It is chosen by you. Two people can build opposite datasets from identical prices.

3

"A 2% target works for every stock"

2% is a rounding error in one name and a huge move in another. Scale by volatility.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 Six trades, four won. What is precision?

67%

2 Which score is useless when defaults are 3%?

accuracy

3 What is the third barrier in the triple barrier?

time

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It replaces the calendar with three finish lines

Target, stop, and a time limit — whichever the price touches first becomes the label.

2

It adds a second model on top

Meta-labeling: the first model says which way, the second says whether to act at all.

3

It ends in code

The problem set labels one trade by hand, then turns the knobs and watches F1 move.

Next: Unit 3 lecture — Labeling and Meta-Labeling